Def. If $s: X \to [0,\infty)$ is a measurable simple function with representation

$$s = \sum_{i=1}^{n} a_i \cdot \chi_{A_i} \quad \text{where} \quad \text{range}(s) = \{a_1, \cdots, a_n\}.$$

For each $E \in \mathcal{M}$, define

$$\int_E s \, d\mu \overset{\text{def}}{=} \sum_{i=1}^{n} a_i \cdot \mu(A_i \cap E).$$

For a measurable function $f: X \to [0,\infty]$, and $E \in \mathcal{M}$, define

$$\int_E f \, d\mu \overset{\text{def}}{=} \sup \left\{ \int_E s \, d\mu \;\middle|\; s \text{ is a simple function s.t } 0 \le s \le f \right\}.$$

This integral is called a Lebesgue integral of f over E, with respect to the measure μ.

Prop. Let $(X, \mathcal{M}, \mu)$ be a measure space, let $f, g: X \to [0,\infty]$ be measurable.

1). If $f \le g$, then $\int_E f \, d\mu \le \int_E g \, d\mu$.

2) If $A \subset B$, then $\int_A f \, d\mu \le \int_B f \, d\mu$

3). If $c \in [0,\infty)$, then $\int_E c \cdot f \, d\mu = c \cdot \int_E f \, d\mu$.

4). If $f|_E \equiv 0$, then $\int_E f \, d\mu = 0$.

5). If $\mu(E) = 0$, then $\int_E f \, d\mu = 0$.

6). $\int_E f \, d\mu = \int_X \chi_E \cdot f \, d\mu$.

Prop. Let s be nonnegative measurable simple function on X.
For each $E \in \mathcal{M}$, define $\varphi(E) \overset{\text{def}}{=} \int_E s\, d\mu$.

Then, the map $\varphi: \mathcal{M} \to [0,\infty]$ is a measure on $(X, \mathcal{M})$ ⌋

Proof. Let $E_n \in \mathcal{M}$ $(n \in \mathbb{N})$ be pairwise disjoints.
Then, for $\mathrm{Im}(s) = \{a_1, \cdots, a_n\}$ and $s^{-1}[\{a_i\}] = A_i$,

$$\varphi\left(\bigcup_{n\in\mathbb{N}} E_n\right) \overset{\text{def}}{=} \sum_{i=1}^{n} a_i \cdot \mu\left(A_i \cap \left(\bigcup_{n\in\mathbb{N}} E_n\right)\right) \qquad A_i \cap \bigcup E_n = \bigcup A_i \cap E_n$$

$$= \sum_{i=1}^{n} a_i \cdot \sum_{r=1}^{\infty} \mu(A_i \cap E_r) \qquad \swarrow \mu \text{ is countably additive}$$

$$= \sum_{r=1}^{\infty} \sum_{i=1}^{n} a_i \cdot \mu(A_i \cap E_r) \qquad \swarrow \text{lemma}$$

$$= \sum_{r=1}^{\infty} \varphi(E_r)$$

And, $\varphi(\emptyset) = 0$, so that φ satisfies definitions of measure. ☑

Lemma. Let $\{s_n\}, \{t_n\}$ be increasing sequences in $[0,\infty]$. Then,

$$\lim_{n\to\infty}(s_n + t_n) = \left(\lim_{n\to\infty} s_n\right) + \left(\lim_{n\to\infty} t_n\right)$$

Proof. Since both are increasing,

$$\lim_{n\to\infty} s_n = \sup_{n\in\mathbb{N}} s_n \quad \text{and} \quad \lim_{n\to\infty} t_n = \sup_{n\in\mathbb{N}} t_n \quad \text{and} \quad \lim_{n\to\infty}(s_n+t_n) = \sup_{n\in\mathbb{N}}(s_n+t_n)$$

And, by the monotonicity, $\geq$ always, $\leq$ by mono.

$$\sup_{n\in\mathbb{N}} s_n + \sup_{n\in\mathbb{N}} t_n = \sup_{n,m\in\mathbb{N}}(s_n + t_m) = \sup_{n\in\mathbb{N}}(s_n + t_n), \text{ so proved.}$$

(For any $N, M \in \mathbb{N}$, take $K = \max\{N, M\}$. Then, by the monotonicity

$$s_N + t_M \leq s_K + t_K \leq \sup_{n\in\mathbb{N}}(s_n + t_n),$$

so $\sup_{n,m}(s_n + t_m) \leq \sup_n (s_n + t_n)$.

Debric.
(Given $\varepsilon > 0$, $\sup_{n,m}(s_n+t_m) - \varepsilon < s_N + t_M \leq \sup_{n,m}(s_n+t_m)$ for some $N, M \in \mathbb{N}$.
Take $K = \max\{N, M\}$. Then, $s_N + t_M \leq s_K + t_K \leq \sup_n (s_n + t_n)$. That is,

$$\sup_{n,m\in\mathbb{N}}(s_n + t_m) - \varepsilon \leq \sup_{n\in\mathbb{N}}(s_n + t_n).$$

Since $\varepsilon > 0$ was arbitrary, we obtain the inequality.) ☑
이거 빼도돼. $s_n = n$, $t_n = 0$ 이라고 $\sup s_n + t_m = \infty$